

AUTONOMOUS LINEFOLLOWING RESCUE ROBOT WITH COMPUTER VISION SYSTEM

Team

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Institution: HTBLA Zeltweg
Higher Technical College for Mechanical Engineering
Country: Austria AT
League: RoboCupJunior Rescue Line



Team Awards

3x National Champion in Austria 🏆🏆🏆
1x National Vice-Champion in Austria
Successful Participation WC 2023 Bordeaux
Successful Participation EC 2024 Hannover
Successful Participation EC 2025 Bari



2023 2024 2025 2026

Role Distribution

May Roman: Embedded Systems & Control: Responsible for the time firmware architecture, sensor fusion algorithms, and low-level motorcontrol loops

Kienberger Markus: Mechanical Design & Simulation: Responsible for parametric master-skeleton modeling (PTC Creo), generative topology optimization, and finite element method (FEM) structural analysis.

Brüggemann Johannes: Computer Vision & Image Processing: Responsible for the optical trackingsystem, HSV color calibration under dynamic lighting, and stochastic object detection.



Markus

Roman

Johannes

Mechanical & Electrical Design

Components

Sensors

- 2x **coloursensor GY31** for detecting green squares/intersections, red for stopping/exit bonus and silver and black/trigger evacuation zone occupation
- 4x **TOF Sensor VL53L7CX** for occupation map (navigation in rescue zone and obstacle avoidance); ball confirmation
- 1x **I2C linearray Sparkfun** for line location for PID-controller
- 1x **ultrasonic sensor HC-SR04** ball and obstacle confirmation (sensorfusion)
- 1x **Camera with SoC Pixy 2** ball and triangle detection

Processors MCUs

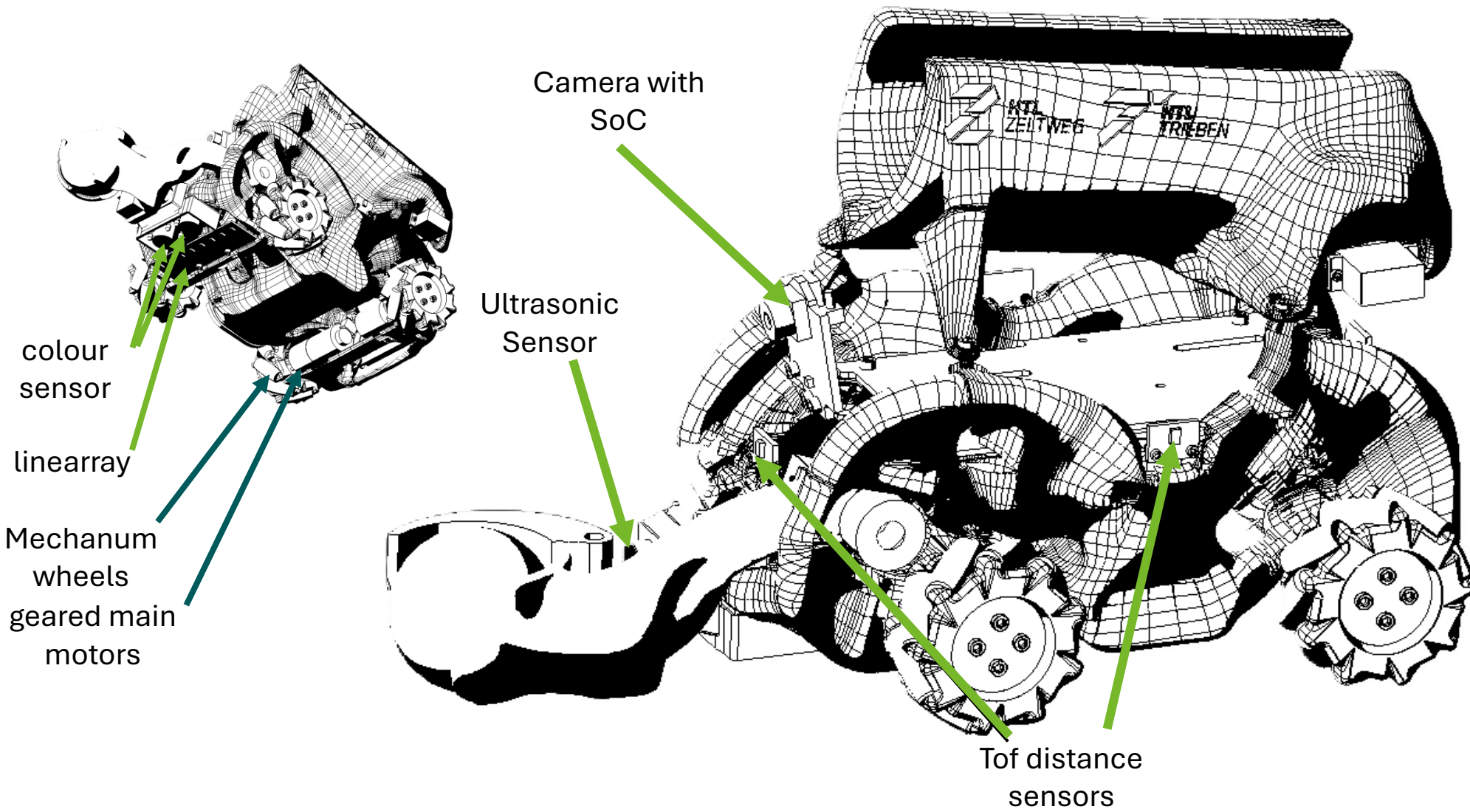
- 1x **ESP 32 DevKit 30 Pin** – as motorcontroller runs PID loop with commands of main Controller
- 1x **ESP 32 S3 DevKit** - as maincontroller runs complete Logic

Actors

- 4x **motor driver DRV8833** for driving main driving motors
- 4x **geared motor with encoder FIT0521 6V 100RPM** main driving motors
- 3x **servo MG90s** for Claw Lifting and Unloading mechanism
- 1x **SPI LCD display 2,4" modul ILI9341 240x320** for debugging and current state

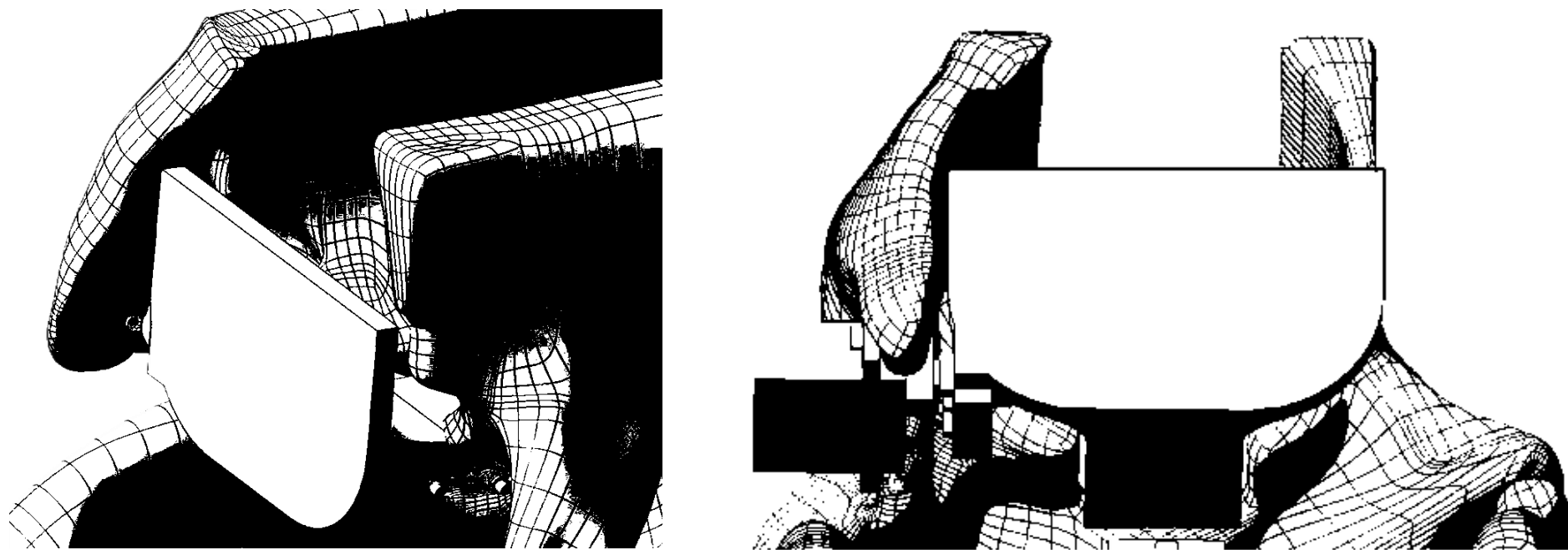
Mechanics

- 4x **Mechanum Wheel 65mm with Lego Hub** driving Wheels
- Fused filament fabricated PETG topology optimized parts** – chassis; ball storage; lifting arm; wheelcouplings these were simulated with crashloads of the Robot to insure mechanical integrity



Intricate Mechanism

Our **ball offloading and seperation mechanism** is one part with only one servo. This enhances reliability and keeps complexity and flimsiness to a minimum



Software Design

Software architecture

The robot's firmware is written in highly optimized C/C++, guaranteeing minimal microcontroller cycle times for instantaneous sensor polling.

The software architecture controls behavior through clearly defined, interrupt-driven states to eliminate unpredictable system lockups (deadlocks) during a run:

- State 1: Line Following → Closed-loop trajectory tracking using the floor sensor array.
- State 2: Obstacle Avoidance → Triggered by ToF thresholds; computes a precise circular bypass trajectory.
- State 3: Intersection Logic → Validates chromatic green markers to execute correct directional turns.
- State 4: Rescue Zone → Transition to primary camera-guided navigation and target hunting.

the robot knows where it is at all times. it knows this because it knows where it isn't. by subtracting where it is from where it isn't, or where it isn't from where it is (whichever is greater), it obtains a difference, or deviation. the guidance subsystem uses deviations to generate corrective commands to drive the robot from a position where it is to a position where it isn't, and arriving at a position where it wasn't, it now is.

Advanced Algorithmic Implementations

HSV Color Space Transformation & Segmentation

To make the image processing immune to fluctuating ambient light (venue spotlights, shadows), RGB camera frames are transformed into the HSV (Hue, Saturation, Value) color space.

Methodology: Static thresholding on the Hue channel isolates the victims (balls) into a binary mask. Centroid extraction algorithms then compute the mathematical vector coordinates of the target, feeding them into the motion planner for omnidirectional alignment and collection.

Empirical Testing & Validation

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All subsystems were subjected to rigorous empirical testing:

- Traction & Incline Performance: The 4WD configuration successfully climbed the maximum regulated 25° ramps with a negligible wheel slip of <3%.
- Computer Vision Robustness: The HSV filtering pipeline achieved a verified victim detection accuracy rate of 98.4% under shifting ambient light levels ranging from 200 Lux to 1200 Lux.
- Controller Optimization: Compared to a baseline P-controller, the fully tuned PID loop reduced path-tracking error in acute corners by 42%, significantly increasing the robot's average course speed.

